Machine Learning 2 — Lecture 1: Course Introduction & Linear Algebra Foundations

Machine Learning 2 — Week 1

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What's on the Menu: Course Content

The course explores advanced machine learning models beyond linear and tree-based methods. Topics include:

- **Introduction:** Limitations of current models and the need for non-linear, high-capacity alternatives.
- **Linear SVM:** Margin theory, primal/dual forms, soft-margin SVM.
- **Kernel Methods:** Kernel trick, Mercer's theorem, various kernels, kernelized SVM/regression.
- **Optimization:** Optimization fundamentals, gradient descent variants, adaptive learning rates, convergence, and constrained optimization.
- **Neural Networks:** Perceptrons, MLPs, activation functions, backpropagation, loss functions, regularization, autoencoders, and CNNs.

> **Note for Review:** Notice the course builds bottom-up — margin theory before kernels, kernels before general optimization, and optimization before neural networks. Each topic is really just "how do we draw a good boundary" answered with progressively more powerful tools.

Grades and Ground Rules: Evaluation & Ethics

Your final grade is split across three components:

Component	Weight
Major Exams	70%
Quiz	10%
Programming Assignment	20%

Together these add up to the full 100% of the course weight.

Ethics: Copying from classmates or the internet, in any form of evaluation, will lead to severe consequences — so do your own SVM derivations.

Vectors 101: The Basics

Before diving into the actual course topics, it's worth spending some time on the linear algebra foundations that will be used throughout — vectors, norms, inner products, distances, and orthonormal matrices all show up repeatedly once you get to margins, kernels, and gradients.

Let $\mathbf{u} = [u_1 \ \cdots \ u_n]^\top$ and $\mathbf{v} = [v_1 \ \cdots \ v_n]^\top$ be two vectors in R^n. As a concrete running example, take $n = 3$:

$$\mathbf{u} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \qquad \mathbf{v} = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$$

Geometrically, if you plot $\mathbf{u}$ and $\mathbf{v}$ as arrows from the origin, they point in different directions and generally have different magnitudes (lengths). The angle between them is denoted θ. This picture — two arrows from a common origin, separated by an angle θ — is the mental model behind every quantity defined below.

Taking the transpose of $\mathbf{u}$ turns the column vector into a row vector:

$$\mathbf{u}^\top = \begin{bmatrix} 1 & 0 & 1 \end{bmatrix}_{1\times 3}$$

The 2-Norm: How Long Is Your Arrow?

The **2-norm** of a vector measures its length — the same notion of "magnitude of a vector" from school, just written with a new symbol: two vertical bars with a subscript 2. By definition, it's the square root of the sum of the squared entries:

$$\|\mathbf{v}\|_2 = \sqrt{v_1^2 + v_2^2 + \cdots + v_n^2}$$

For the example vector $\mathbf{u} = [1, 0, 1]^\top$:

$$\|\mathbf{u}\|_2 = \sqrt{1^2 + 0^2 + 1^2} = \sqrt{2}$$

A useful related identity is that the inner product of a vector with itself equals the square of its 2-norm:

$$\mathbf{v}^\top \mathbf{v} = v_1^2 + v_2^2 + \cdots + v_n^2 = \|\mathbf{v}\|_2^2$$

This makes sense once you see the inner product defined below — taking the inner product of $\mathbf{u}$ with itself just sums the squares of its own entries, which is exactly what sits under the square root in the norm.

The Inner Product: Vectors Shaking Hands

The **inner product** (also called the dot product, as seen in school) of two vectors $\mathbf{u}, \mathbf{v} \in \mathrm{R}^n$ is defined as:

$$\langle \mathbf{u}, \mathbf{v} \rangle = \mathbf{u}^\top \mathbf{v} = \mathbf{v}^\top \mathbf{u} = \langle \mathbf{v}, \mathbf{u} \rangle$$

Concretely, this is the matrix product of the row vector $\mathbf{u}^\top$ with the column vector $\mathbf{v}$, which works out to the sum of the element-wise products of the two vectors' entries:

$$\langle \mathbf{u}, \mathbf{v} \rangle = u_1 v_1 + u_2 v_2 + \cdots + u_n v_n$$

The order doesn't matter — $\mathbf{u}^\top \mathbf{v}$ and $\mathbf{v}^\top \mathbf{u}$ give the same scalar.

Working through the example vectors: $\mathbf{u}^\top = [1 \ 0 \ 1]$ and $\mathbf{v} = [0, 1, 1]^\top$ (note $\mathbf{v}$ is left as a column here, not transposed), so:

$$\langle \mathbf{u}, \mathbf{v} \rangle = \mathbf{u}^\top \mathbf{v} = \begin{bmatrix} 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} = 1 \times 0 + 0 \times 1 + 1 \times 1 = 1$$

There is also a geometric definition of the inner product, connecting it back to the angle θ between the two vectors:

$$\langle \mathbf{u}, \mathbf{v} \rangle = \|\mathbf{u}\|_2 \times \|\mathbf{v}\|_2 \cos\theta$$

So, given two vectors, you can relate their inner product directly to their individual lengths and the angle between them — this relationship becomes the basis for the definition of orthogonality later.

Distance: How Far Apart Are Two Vectors?

The **distance** between two vectors $\mathbf{u}$ and $\mathbf{v}$ is the length (2-norm) of the vector joining them. If $\mathbf{e} = \mathbf{v} - \mathbf{u}$ is that joining vector, then the distance is simply $\|\mathbf{e}\|_2$:

$$\|\mathbf{u} - \mathbf{v}\|_2 = \sqrt{(\mathbf{u} - \mathbf{v})^\top (\mathbf{u} - \mathbf{v})} = \sqrt{\mathbf{u}^\top \mathbf{u} - 2\mathbf{u}^\top \mathbf{v} + \mathbf{v}^\top \mathbf{v}}$$

Written entry-wise:

$$\|\mathbf{u} - \mathbf{v}\|_2 = \sqrt{(u_1 - v_1)^2 + \cdots + (u_n - v_n)^2}$$

For the running example, $\mathbf{e} = \mathbf{v} - \mathbf{u}$, and:

$$\|\mathbf{e}\|_2 = \sqrt{(1-0)^2 + (0-1)^2 + (1-1)^2} = \sqrt{2}$$

> **Note for Review:** Distance is one of those quietly load-bearing concepts you'll meet constantly later in the course. When training a neural network, the loss function typically compares a **predicted output vector** against the **ground-truth vector** — and that comparison is, at its core, a distance calculation. A large distance means a large error, which is exactly the signal used to update the network's weights during training.

Orthogonal Vectors: Strictly 90°

Two vectors $\mathbf{u}, \mathbf{v} \in \mathrm{R}^n$ are called **orthogonal** if the angle θ between them is exactly $90°$. Since $\cos 90° = 0$, the geometric definition of the inner product tells us this is equivalent to a much more practical test:

$$\mathbf{u}, \mathbf{v} \text{ are orthogonal} \iff \langle \mathbf{u}, \mathbf{v} \rangle = \mathbf{u}^\top \mathbf{v} = 0$$

So checking orthogonality never actually requires measuring an angle — just compute the inner product and see if it's zero.

Two unit-norm vectors $\mathbf{u}, \mathbf{v} \in \mathrm{R}^n$ (i.e., vectors whose 2-norm equals 1) are orthogonal to each other exactly when $\langle \mathbf{u}, \mathbf{v} \rangle = 0$.

Worked example — an orthogonal pair. Take:

$$\mathbf{u} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, \qquad \mathbf{v} = \frac{1}{\sqrt{2}} \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

Both are scaled by $\frac{1}{\sqrt{2}}$ precisely so that each has unit norm. Their inner product is:

$$\langle \mathbf{u}, \mathbf{v} \rangle = \mathbf{u}^\top \mathbf{v} = \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} + 0 \times 0 = \frac{1}{2} - \frac{1}{2} = 0$$

Since the inner product is zero, $\mathbf{u}$ and $\mathbf{v}$ are orthogonal.

Worked example — a non-orthogonal pair. Now take:

$$\mathbf{u} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \qquad \mathbf{v} = \frac{1}{\sqrt{2}} \begin{bmatrix} -1 \\ -1 \end{bmatrix}$$

Both are unit-norm vectors, but:

$$\mathbf{u}^\top \mathbf{v} = \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} = \frac{2}{\sqrt{2}} \neq 0$$

so this pair is **not** orthogonal, even though both vectors have unit length. Unit norm alone says nothing about direction — orthogonality is entirely about the inner product being zero.

Orthonormal Matrices: The Well-Behaved Ones

A matrix is naturally viewed as a collection of column vectors stacked side by side. For example, a 2×2 matrix $\mathbf{A}$ and a 2×3 matrix $\mathbf{B}$ can be written entrywise:

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}_{2\times 2}, \qquad \mathbf{B} = \begin{bmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \end{bmatrix}_{2\times 3}$$

or, in column notation, as $\mathbf{A} = [\mathbf{a}_1 \ \mathbf{a}_2]$ with $\mathbf{a}_1, \mathbf{a}_2 \in \mathrm{R}^{2\times 1}$, and $\mathbf{B} = [\mathbf{b}_1 \ \mathbf{b}_2 \ \mathbf{b}_3]$ with $\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3 \in \mathrm{R}^{2\times 1}$.

Comparing two 2×2 matrices. Consider:

$$\mathbf{A} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix} = [\mathbf{a}_1 \ \mathbf{a}_2], \qquad \mathbf{B} = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix} = [\mathbf{b}_1 \ \mathbf{b}_2]$$

For $\mathbf{A}$: $\|\mathbf{a}_1\|_2 = \sqrt{2}$ and $\|\mathbf{a}_2\|_2 = \sqrt{5}$ — the columns don't even have equal length, let alone unit length, and $\mathbf{a}_1^\top \mathbf{a}_2 \neq 0$.

For $\mathbf{B}$: both columns have unit length, $\|\mathbf{b}_1\|_2 = \|\mathbf{b}_2\|_2 = 1$, and they are orthogonal to each other, $\mathbf{b}_1^\top \mathbf{b}_2 = 0$. This combination — unit-length columns that are pairwise orthogonal — is what makes $\mathbf{B}$ special and $\mathbf{A}$ not.

A second example — the identity matrix. Take the 3×3 identity:

$$\mathbf{C} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = [\mathbf{c}_1 \ \mathbf{c}_2 \ \mathbf{c}_3]$$

Here, $\|\mathbf{c}_1\|_2 = \|\mathbf{c}_2\|_2 = \|\mathbf{c}_3\|_2 = 1$ (unit length), and every pair of distinct columns is orthogonal: $\mathbf{c}_1^\top \mathbf{c}_2 = 0$, $\mathbf{c}_2^\top \mathbf{c}_3 = 0$, $\mathbf{c}_3^\top \mathbf{c}_1 = 0$.

Formal definition. Whenever a *square* matrix's columns all have unit length **and** are pairwise orthogonal, the matrix is called an **orthonormal matrix** (also called an orthogonal matrix). Formally, for a matrix $\mathbf{A}$ with n columns $\mathbf{a}_1, \ldots, \mathbf{a}_n \in \mathrm{R}^{n\times 1}$, $\mathbf{A}$ is orthonormal if:

$$\langle \mathbf{a}_i, \mathbf{a}_j \rangle = 1 \text{ if } i = j, \qquad \langle \mathbf{a}_i, \mathbf{a}_j \rangle = 0 \text{ if } i \neq j$$

The first condition ($i = j$) is really just saying each column has unit norm, since the inner product of a vector with itself is the square of its 2-norm. The second condition ($i \neq j$) is exactly the orthogonality condition from before, applied to every pair of columns.

Key consequence: If $\mathbf{A}$ is an orthonormal matrix, then $\mathbf{A}^\top \mathbf{A} = \mathbf{I}$ (the identity matrix). This can be verified directly for a 3×3 matrix $\mathbf{A} = [\mathbf{a}_1 \ \mathbf{a}_2 \ \mathbf{a}_3]$. Taking the transpose turns each column into a row:

$$\mathbf{A}^\top = \begin{bmatrix} \mathbf{a}_1^\top \\ \mathbf{a}_2^\top \\ \mathbf{a}_3^\top \end{bmatrix}$$

Multiplying $\mathbf{A}^\top$ by $\mathbf{A}$ means taking every row of $\mathbf{A}^\top$ (i.e., every column of $\mathbf{A}$) against every column of $\mathbf{A}$:

$$\mathbf{A}^\top \mathbf{A} = \begin{bmatrix} \mathbf{a}_1^\top \\ \mathbf{a}_2^\top \\ \mathbf{a}_3^\top \end{bmatrix} [\mathbf{a}_1 \quad \mathbf{a}_2 \quad \mathbf{a}_3] = \begin{bmatrix} \mathbf{a}_1^\top \mathbf{a}_1 & \mathbf{a}_1^\top \mathbf{a}_2 & \mathbf{a}_1^\top \mathbf{a}_3 \\ \mathbf{a}_2^\top \mathbf{a}_1 & \mathbf{a}_2^\top \mathbf{a}_2 & \mathbf{a}_2^\top \mathbf{a}_3 \\ \mathbf{a}_3^\top \mathbf{a}_1 & \mathbf{a}_3^\top \mathbf{a}_2 & \mathbf{a}_3^\top \mathbf{a}_3 \end{bmatrix}$$

Since $\mathbf{A}$ is orthonormal, every diagonal entry $\mathbf{a}_i^\top \mathbf{a}_i = 1$ (unit length) and every off-diagonal entry $\mathbf{a}_i^\top \mathbf{a}_j = 0$ for $i \neq j$ (orthogonality). So the whole product collapses to:

$$\mathbf{A}^\top \mathbf{A} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \mathbf{I}$$

> **Note for Review:** This identity, $\mathbf{A}^\top \mathbf{A} = \mathbf{I}$ for orthonormal $\mathbf{A}$, is one of the most reused facts in linear-algebra-heavy ML derivations (PCA, SVD, rotation matrices all lean on it). The intuition to hold onto: orthonormal matrices represent transformations that only rotate or reflect vectors — they never stretch or shrink them — which is precisely why multiplying by $\mathbf{A}^\top$ perfectly "undoes" multiplying by $\mathbf{A}$.